

Overview of SQL Server

What is SQL Server?

Microsoft SQL Server is a relational database management system (RDBMS) developed by Microsoft. It's used to store and retrieve data requested by other software, either on the same computer or over a network.

SQL Server Editions:

Edition	Description
Express	Free, lightweight version, limited features (ideal for learning or small apps).
Standard	Full-featured, supports moderate workloads, more advanced security & performance tools.
Enterprise	Advanced features like partitioning, advanced security, high availability, in-memory processing.

Features of SQL Server 2014:

In-Memory OLTP (Hekaton): Improves performance for transactional workloads.

Buffer Pool Extension: Uses SSDs to extend memory.

Backup Encryption: Encrypt database backups.

AlwaysOn Enhancements: Improved high availability.

SQL Server Architecture:

Relational Engine: Executes queries and manages memory, plan cache.

Storage Engine: Handles disk I/O, reading/writing data pages.

SQL OS: Abstracts operating system functionality (scheduling, memory).

Databases: Collection of files (MDF, NDF, LDF) with objects like tables, views, procedures.

SQL Server Management Studio (SSMS):

Graphical interface to manage SQL Server.

Allows executing queries, managing objects, backups, security, etc.

Installing SQL Server 2014

System Requirements:

OS: Windows 7 or later, Windows Server.

RAM: Minimum 1 GB (4 GB+ recommended).

Disk: At least 6 GB for installation.

.NET Framework: 3.5 or later.

Installation Steps:

Download installer.

Run setup.exe.

Choose New SQL Server stand-alone installation.

Choose features: Database Engine, SSMS, Agent.

Configure authentication (Mixed Mode or Windows Auth).

Install.

Connecting to the Server:

Use SSMS → Enter:

Server Name: localhost or .\SQLEXPRESS

Authentication: SQL Server Auth or Windows Auth

SQL Server Services:

Service	Description
SQL Server (MSSQLSERVER)	Main database engine.
SQL Server Agent	Schedules jobs and automates tasks.
SQL Browser	Allows connections via named instances.

SQL Server Management Studio (SSMS)

SSMS Interface Overview:

Object Explorer: Navigate servers, databases, objects.

Query Editor: Write and run SQL scripts.

Properties Window: View properties of objects.

Creating a Database (GUI & SQL):

Using GUI:

Right-click Databases > New Database.

Enter name > OK.

Using SQL:

```
CREATE DATABASE StudentDB;
```

```
GO
```

Creating Tables (GUI & SQL):

Using GUI:

Expand database > Right-click Tables > New Table.

Add columns with data types > Save as Students.

Using SQL:

```
USE StudentDB;
```

```
GO
```

```
CREATE TABLE Students (
```

```
    StudentID INT PRIMARY KEY,  
    FullName NVARCHAR(100),  
    Age INT,
```

```
EnrollmentDate DATE
```

```
);
```

```
GO
```

Exploring Object Explorer:

Databases > Expand to see Tables, Views, Stored Procedures, Security.

Can also right-click to generate scripts.

Query Editor Basics:

Write and execute queries.

View messages, results, and execution plans.

Example Query:

```
-- Insert data
```

```
INSERT INTO Students (StudentID, FullName, Age, EnrollmentDate)
```

```
VALUES (1, 'Ali Khan', 22, '2023-08-15');
```

```
-- Select data
```

```
SELECT * FROM Students;
```

In []: